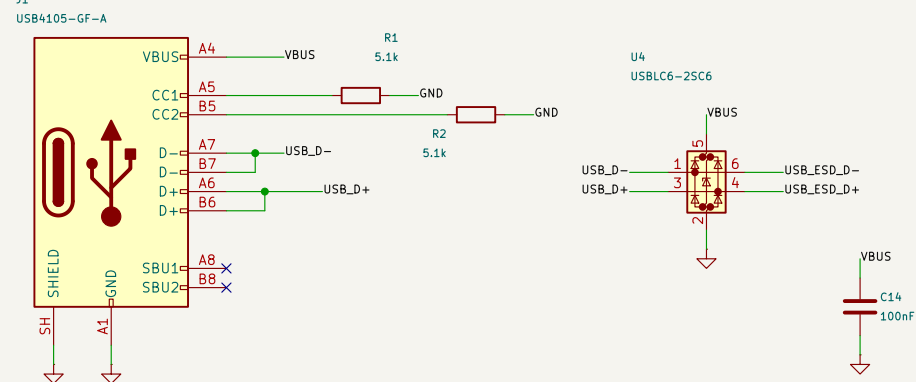


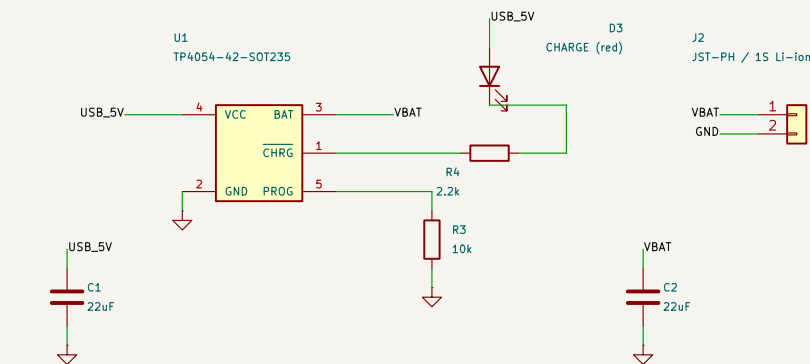
NUCULA v2 / USB-C, Li-ion power and ESP32-C3

01 USB 2.0 device / 5 V sink



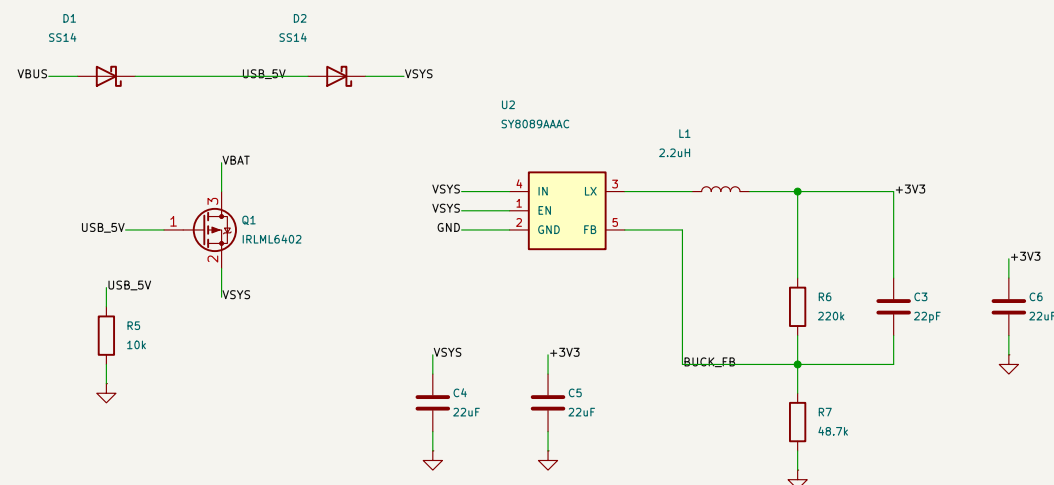
CC1 / CC2: independent 5.1k Rd. No USB-PD.
Route D+/D- as a 90 ohm differential pair.
Place U4 at connector; R12/R13 at module.
See design notes for USB current / inrush limits.

02 4.2 V / 100 mA charger



Protected 1S Li-ion/LiPo only (4.2 V full).
J2 pin 1 = BAT+; pin 2 = GND. Check pack polarity!
100 mA nominal; termination -10 mA.
No cell-temperature input. Confirm battery rating.

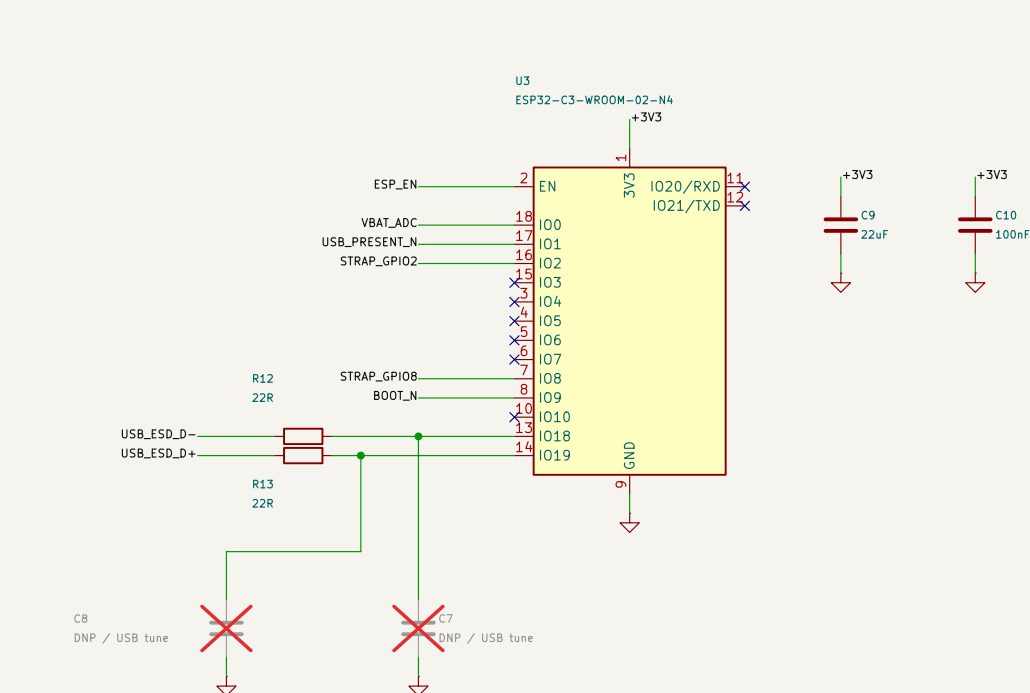
03 USB-priority supply / 3.31 V buck



Q1 drain = battery, source = VSYS.
Body diode points VBAT $\rightarrow$ VSYS.
USB load bypasses charge-current sensing.

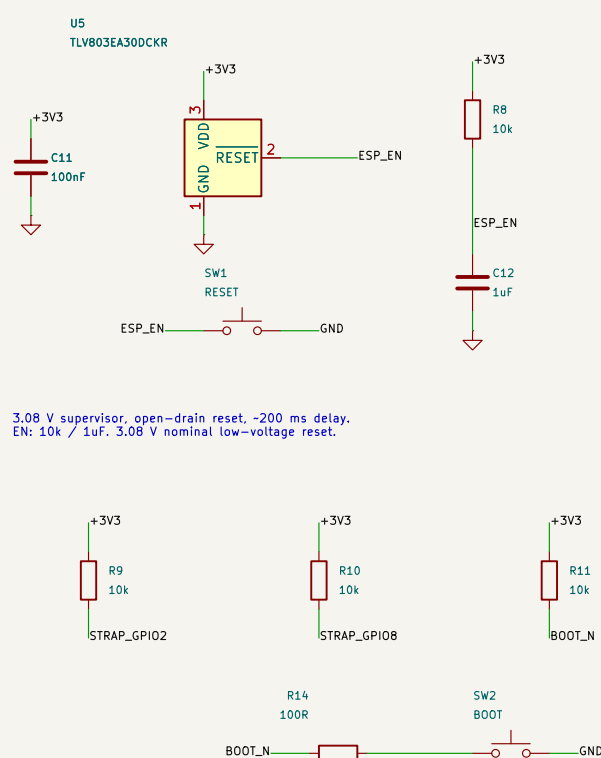
VOUT = 3.31 V; R6 / R7: 0.1% precision
500 mA rail design target; validate transients.
Buck only: low battery causes dropout / reset.
C5 + C6: 44 uF nominal; verify DC-bias derating.

04 ESP32-C3-WROOM-02-N4 / native USB



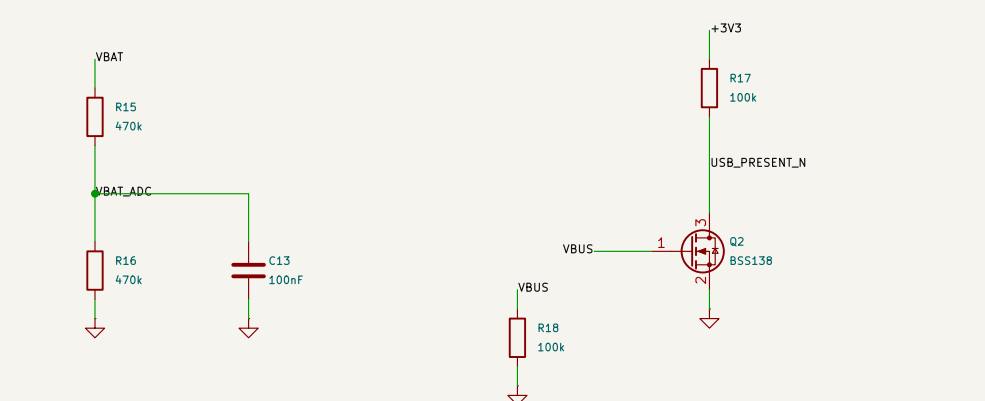
USB D- = GPIO18 (pin 13); D+ = GPIO19 (pin 14).
No UART bridge. Unused GPIOs deliberately marked NC.
Keep antenna clear; bond EP pin 19 and pin 9 to ground.
Use ADC1 GPIO0 for battery sensing (not ADC2).

05 Reset, undervoltage and boot straps



Hold BOOT; tap RESET to enter ROM download mode.
GPIO8 must stay high while GPIO9 is low.
No large capacitor on GPIO9.

06 Battery and USB presence sensing



VBAT_ADC = VBAT / 2, max -2.15 V incl. tolerances.
ADC1 GPIO0: use calibrated 11/12 dB attenuation.
Divider load -4.5 uA; allow >=120 ms to settle.

USB_PRESENT_N: GPIO1 LOW when VBUS is present.
Firmware must disable USB attach when VBUS is absent.
Q2 isolates 5 V VBUS from the 3.3 V GPIO supply.

DRAFT / before PCB:
 Confirm protected battery and JST polarity.
 Validate USB current, inrush, suspend and ESD.
 Verify regulator layout, capacitor bias and load steps.
 Reference: Olimex ESP32-C3-DevKit-Lipo Rev C
 See docs/power-design.md for sources and limits.

Schematic only / protected 4.2 V single-cell Li-ion / 100 mA charge
Adapted from Olimex ESP32-C3-DevKit-Lipo Rev C; CERN-OHL-S-2.0

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Title: Nucula v2 – USB-C, battery power and ESP32-C3

Size: A3

Date: 2026-09-19

Rev: A-draft

Id: 1/1

ERC source declarations